Rezolvări Huffman

1. Toți subarborii unui arbore binar optim sunt de asemenea optimi
R: True

2. Într-un arbore optim nodurile la care accesul se face mai frecvent devin noduri cu pondere mai mare, cele vizitate mai rar, noduri cu pondere mai R: mică.

3. Se numește arbore binar optim, arborele a cărui structură aduce la R: cost minim

4. Se dorește a codifica fiecare caracter...
R: prefix

5. Un cod prefix poate fi reprezentat printr-un arbore binar dacă nodurilor terminale

6. Afirmații adev:
1. Codarea Huffman este folosită pt. compresie
2. În codarea Huffman, niciun cod nu este prefixul altui cod
3. Algoritmii de tip Huffman static au dezavantajul că necesită cunoașterea prealabilă a frecvențelor de apariție pentru fiecare simbol din sursă

Ex.

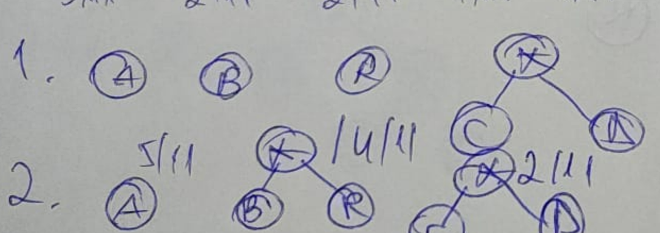
ABRACADABRA

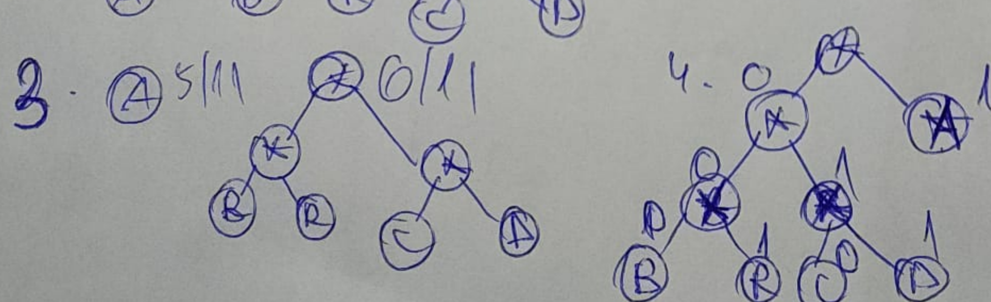
A - $\frac{5}{11}$; B - $\frac{2}{11}$; R - $\frac{2}{11}$, C - $\frac{1}{11}$, D - $\frac{1}{11}$ R: 23 lungime șir 11

A B R C D
5/11 2/11 2/11 1/11 1/11

A = 5*1 = 5
B = 3*2 = 6
C = 3
D = 3
R = 6

5+6+6+6 = 23





INDISTINCTIBILI lungime: 15

$I - \frac{6}{15}$, $N - \frac{2}{15}$, $D - \frac{1}{15}$, $S - \frac{1}{15}$, $T - \frac{2}{15}$, $C - \frac{1}{15}$, $B - \frac{1}{15}$, $L - \frac{1}{15}$

1.

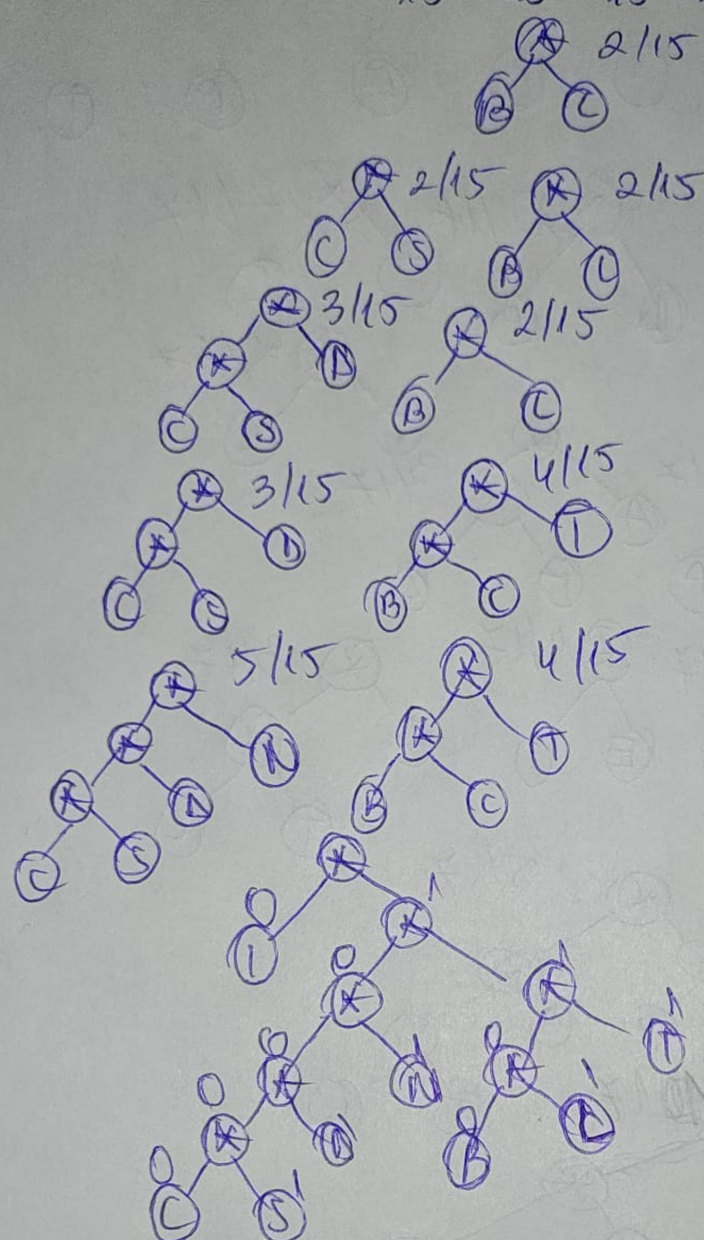
2.

3.

4.

5.

6+7

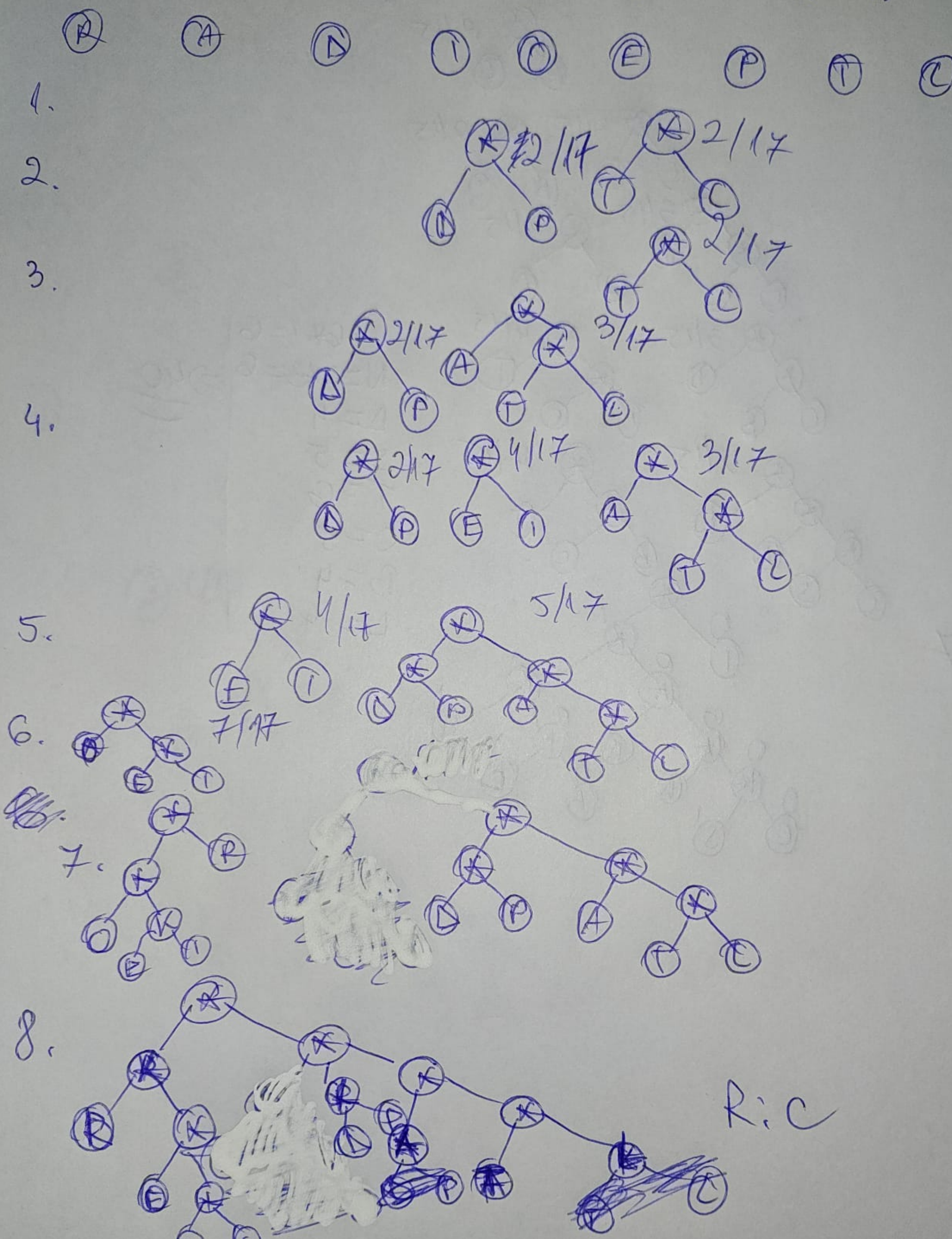


$i = 6 * 1 = 6$
$N = 2 * 3 = 6$
$D = 4$
$S = 5$
$T = 6$
$C = 5$
$B = 4$
$L = 4$

$\Rightarrow \underline{40}$

RADIOREPORTERILOR lungime = 17

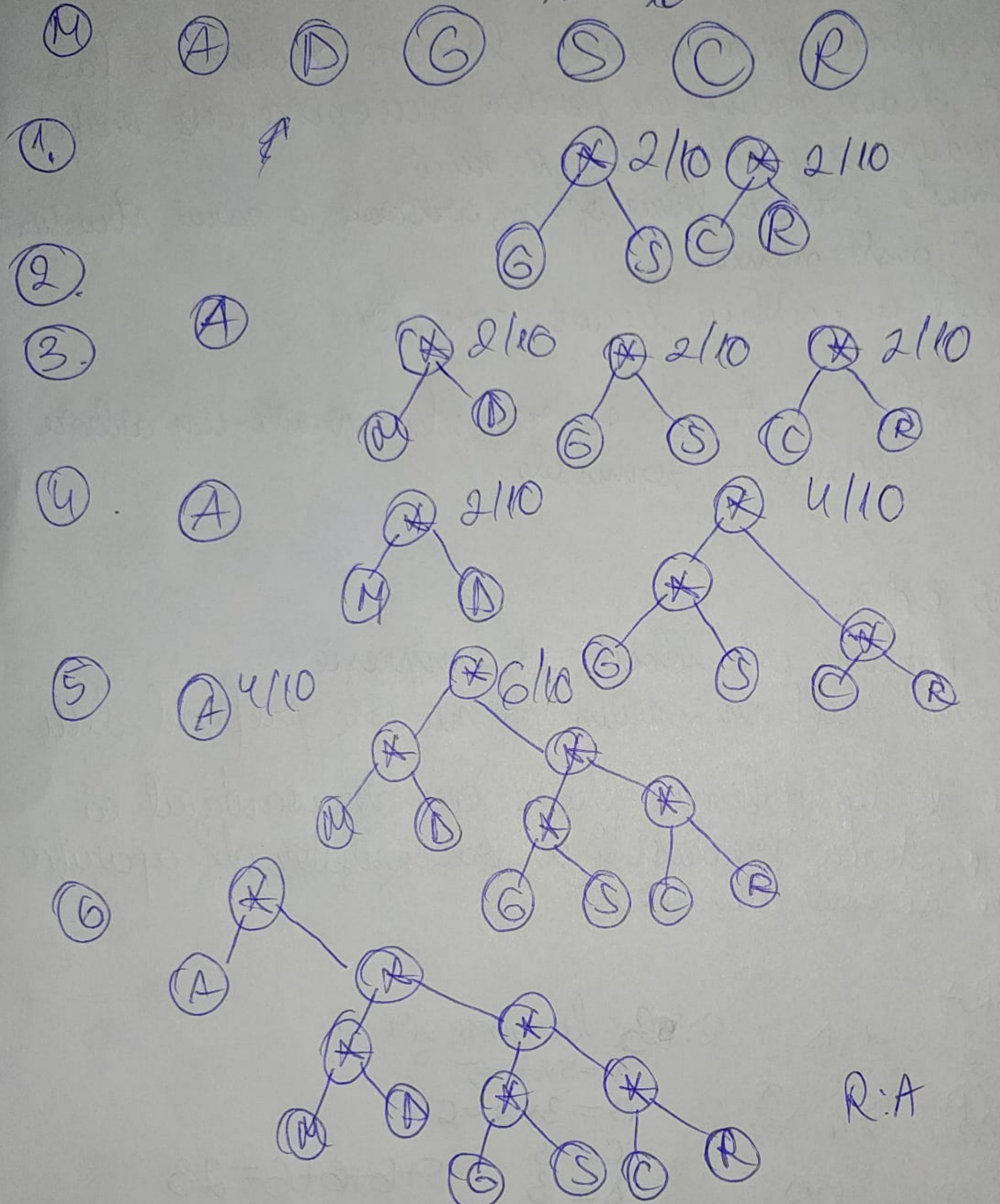
R – $\frac{5}{17}$, A – $\frac{1}{17}$, D – $\frac{1}{17}$, I – $\frac{2}{17}$, O – $\frac{3}{17}$, E – $\frac{2}{17}$, P – $\frac{1}{17}$, T – $\frac{1}{17}$, L – $\frac{1}{17}$



R: C

* MADAGASCAR

M − $\frac{1}{10}$, A − $\frac{4}{10}$, D − $\frac{1}{10}$, G − $\frac{1}{10}$, S − $\frac{1}{10}$, C − $\frac{1}{10}$, R − $\frac{1}{10}$, lungime = 10



R: A

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estion 4
t yet
swered
rked out of
0
Flag question

Se doreşte a se codifica fiecare caracter printr-o secvență de cifre binare "0" şi "1", astfel încât codul unui caracter, să nu fie [ ] pentru codul nici unui alt caracter.

prefix
sufix
duplicat

Me fix

Previous page

Finish attempt ...

ENG 18:1 15.04.20

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Quiz navigation

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Time left **0:18:37**

Question **2**
Not yet answered
Marked out of 2.00
Flag question

Un cod prefix (care are proprietatea de prefix) poate fi reprezentat printr-un arbore binar, dacă nodurilor [terminale] ale arborelui li se asociază caracterele originale ale alfabetului, iar ramurilor

- neterminale
- cu chei pare
- terminale
- cu ponderea cea mai mare

Previous page | Next page

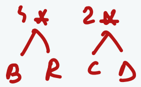
◄ Test AVL | Jump to... | Test Arbori B ►

Course: Proiectarea ... | Algoritmul lui Prim | Algoritmul lui Prim | Facebook | Test Huffman (page 4

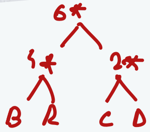
A - 5
B - 2
R - 2
C - 1
D - 1

1.

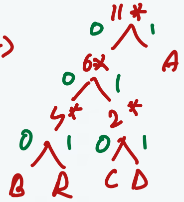
2. A - 5
B - 2
R - 2
C,D - 2 =>



3. A - 5
B,R - 4
C,D - 2 =>



4. A - 5
B,R,C,D - 6 =>



=> A - 1 => 5 * 1
B - 000 3 * 2
R - 001 3 * 2
C - 010 3 * 1
D - 011 3 * 1

=> 5 + 6 + 6 + 3 + 3
= 23

Brad Ana-Alexandra

Vladimir
Razvan

Shortcut

A - 5
B - 2
R - 2 =>
C - 1
D - 1

A - 5
B - 2
R - 2 =>
C,D - 2

A - 5
B,R - 4
C,D - 2 =>

A - 5
B,C,D,R - 6

=> A,B,C,D,R - 11

A	B	R	C	D
1	1	1	1	1
	1	1	1	1
	1	1	1	1
1	3	3	3	3

$\Rightarrow 5\cdot 1 + 3\cdot 2 + 3\cdot 2 + 3 + 3$

$\Rightarrow 5 + 6 + 6 + 6 = 23$

Test Huffman

Numarul de biti necesar codarii cuvantului urmator, folosind coduri Huffman este:

ABRACADABRA

Answer: 23

Previous page

Finish attempt ...

11:28 AM
02-Apr-20

92953091_229626568251918_2138960762142457856_n.png

Question **2**

Answer saved

Marked out of 2.00

Flag question

Intr-un arbore optim nodurile la care accesul se face mai frecvent devin noduri cu pondere mai mare , cele vizitate mai rar, noduri cu pondere mai mica .

RADIO REPORTERILOR

R - 5
A - 1
D - 1
i - 2
O - 3
E - 2
P - 1
T - 1
L - 1

=> 2* (A D) 2* (P T) =>

R - 5
A,D - 2
i - 2
O - 3
E - 2
P,T - 2
L - 1

=> 2* (A D) 3* (2* (P T) L)

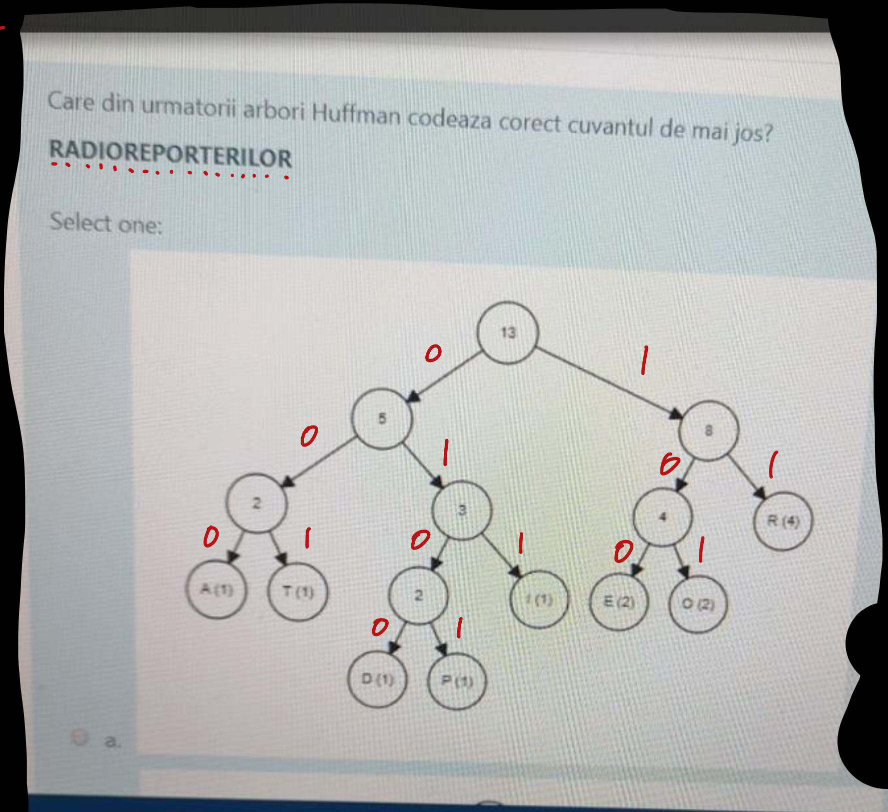
=> R - 5
A,D - 2
i - 2
O - 3
E - 2
P,T,L - 3

=> 4* (i 2* (A D)) 3* (2* (P T) L)

=> R - 5
A,D,i - 4
O - 3 } 7
E - 2
P,T,L - 3 } 5

=> 7* (4* (i 2* (A D)) O) 5* (3* (2* (P T) 2) 2)

=> 10* (R 5*)



R 11
A 000
D 0100
i 011
O 101
E 100
P 0101
T 001
L

11 000

Finish attempt ...

Time left **0:14:02**

Question 1

Not yet answered

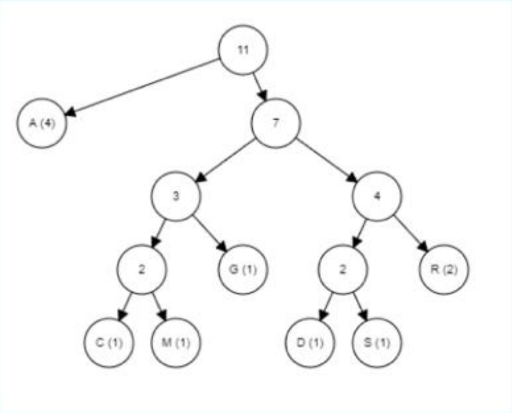
Marked out of 3.00

Flag question

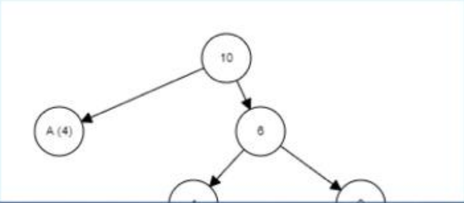
Care din urmatorii arbori Huffman codeaza corect cuvantul de mai jos?

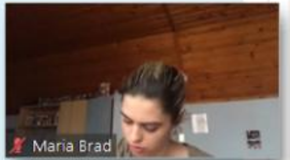
MADAGASCAR

Select one:



a.





M - 1
A - 4
D - 1
G - 1
S - 1
C - 1
R - 1

se vede din frecventa

Test Huffman (page 3 of 4) | Post Attendee - Zoom

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Quiz navigation

Finish attempt ...

Time left **0:18:20**

Question **3**
Not yet answered
Marked out of 2.00
Flag question

Toți subarborii unui arbore binar optim sunt de asemenea optimi.

Select one:

0 True

False

Previous page | Next page

◄ Test AVL | Jump to... | Test Arbori B ►

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10:16 AM
4/2/2020

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e Laborator ▸ Test Huffman

estion 1
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arked out of
00
Flag question

Se numeşte arbore binar optim, arborele a cărui structură conduce la

un cost minim

✓

◄ Test AVL

Jump to...

Te

You are logged in as [illegible]

+4
More

O2-PAA ▸ Teste Laborator ▸ Test Huffman

Question 3
Answer saved
Marked out of 2.00
Flag question

Care din urmatoarele afirmatii despre codurile Huffman este adevarata:

Select one or more:

- [ ] a. Algoritmii de tip Huffman static au dezavantajul că necesită cunoaşterea prealabilă a frecventelor de aparitie pentru fiecare simbol din sursa.
- [ ] b. Codarea Huffman poate prezenta pierderi (de informatie), in anumite cazuri.
- [x] c. Codarea Huffman este folosita pentru compresie.
- [x] d. In codarea Huffman, niciun cod nu este prefixul altui cod.

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Cum se face deplasarea intr-un arbore Huffman?

- a. 0 = nu se face deplasare; 1 = in adancime
- b. 0 = stanga; 1 = dreapta
- c. 1 = stanga; 2 = dreapta
- d. 0 = dreapta; 1 = stanga

Jump to...

Care din urmatoarele afirmatii despre codurile Huffman este adevarata:

Alegeți una sau mai multe opțiuni:

- [x] a. Algoritmii de tip Huffman static au dezavantajul că necesită cunoașterea prealabilă a frecventelor de aparitie pentru fiecare simbol din sursa.
- [x] b. In codarea Huffman, niciun cod nu este prefixul altui cod.
- [x] c. Codarea Huffman este folosita pentru compresie.
- [ ] d. Codarea Huffman poate prezenta pierderi (de informatie), in anumite cazuri.

a precedentă

Question **2**

Answer saved

Marked out of 3.00

Flag question

Care din urmatorii arbori Huffman codeaza corect cuvantul de mai jos?

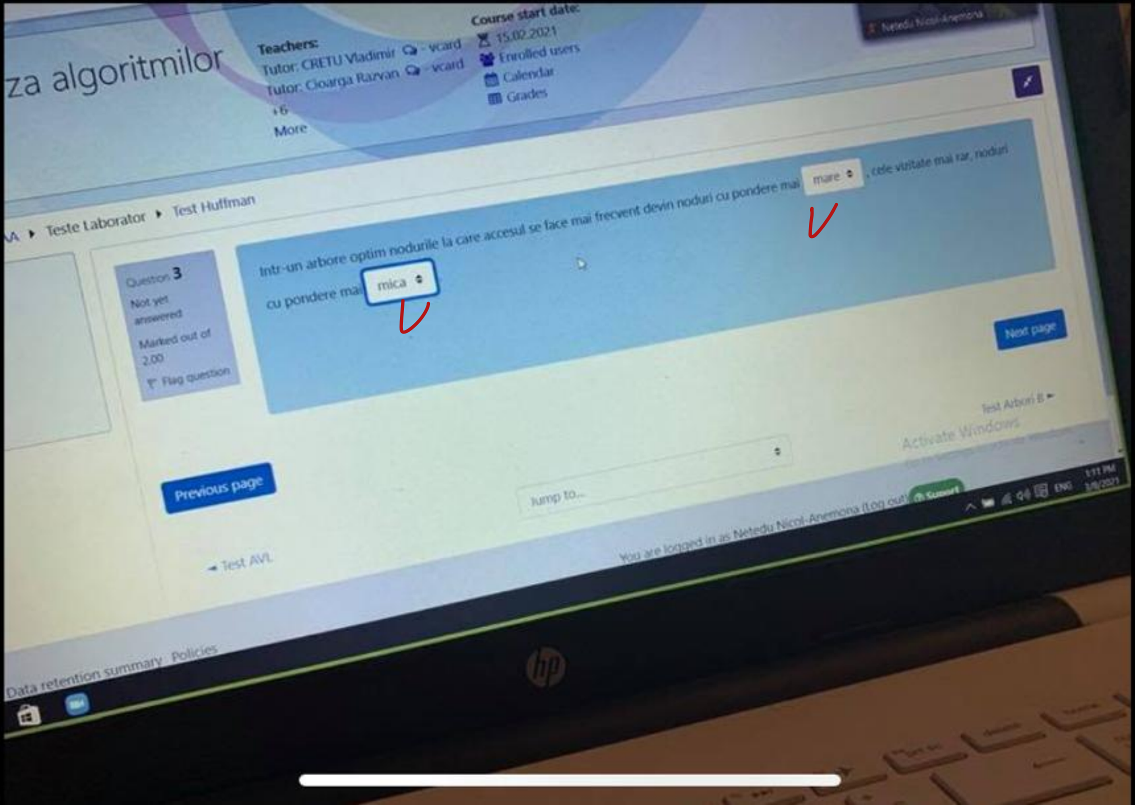
COTCODACIT

???

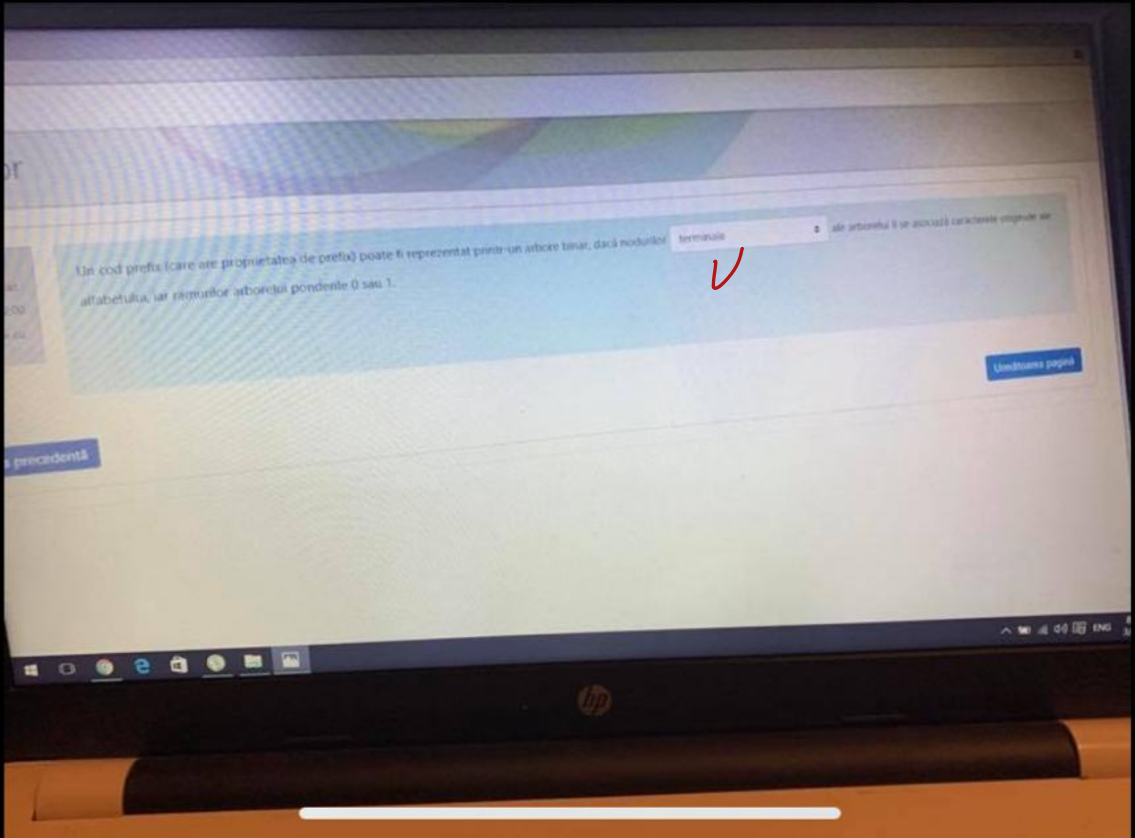
Select one:

a. 

198832308_561719616168843_4_2384328217136375793_n.jpg



198808361_1211818959257575_4782059876779159046_n.jpg



A - 6
G - 1
R - 1
M - 1
T - 3
i - 2
C - 1
L - 1
E - 1

=> A - 6
G,R - 2
M,C - 2
T - 3
i - 2
LE - 2

=> A - 6
G,R,M,C - 4
T - 3
LE,i - 4

=> A - 6
G,R,M,C - 4
T,L,E,i - 7

=> A,G,R,M,C - 10
T,E,E,i - 7

A	G	R	M	T	i	C	L	E
2	4	4	4	2	3	4	4	4

=> 2*6 + 4 + 4 + 4 + 3*2 + 2*3 + 12 = 12 + 12 + 6 + 6 + 12 = 48

Numarul de biti necesar codarii cuvantului urmator, folosind coduri Huffman este:

AGRAMATICALITATEA

Răspuns: 48

dentă

194939998_3195497930585260_7884604938981409020_n.png

Teste Laborator ▸ Test Huffman

Quiz

Finish attempt ...

Time left **0:19:19**

Question **3**

Not yet answered

Marked out of 3.00

Flag question

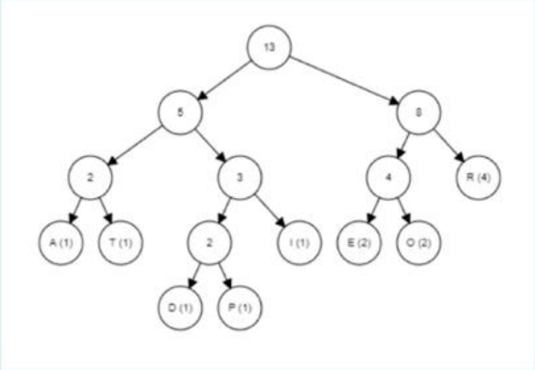
Care din urmatorii arbori Huffman codeaza corect cuvantul de mai jos?

RADIOREPORTERILOR => L = 17

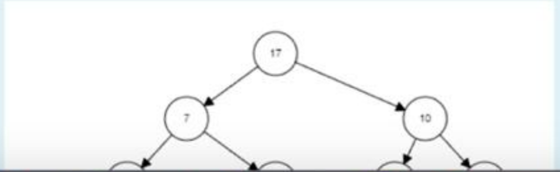
R - 5
A - 1
D - 1
i - 2
O - 3
E - 2
P - 1
T - 1
L - 1

Select one:

X



a.



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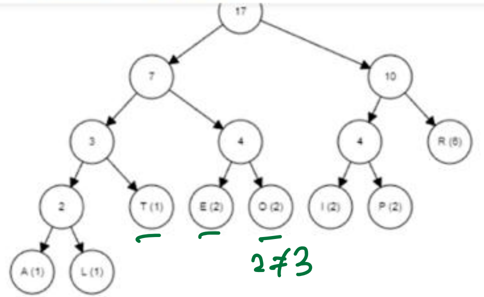
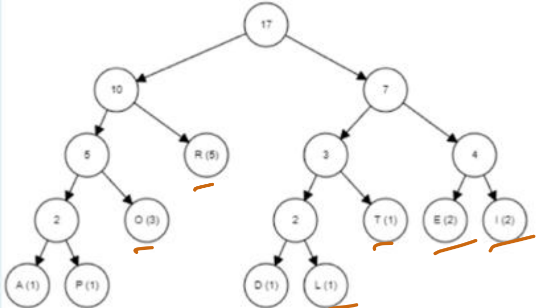


2≠3

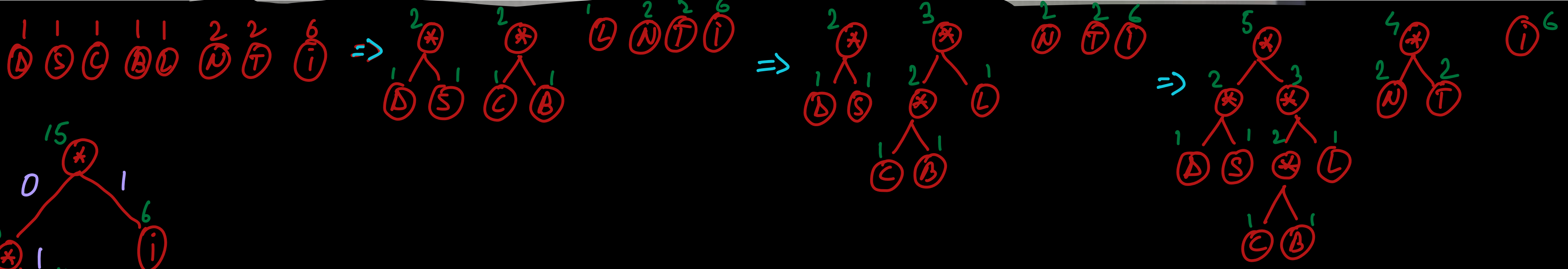
R - 5
A - 1
D - 1
I - 2
O - 3
E - 2
P - 1
T - 1
L - 1

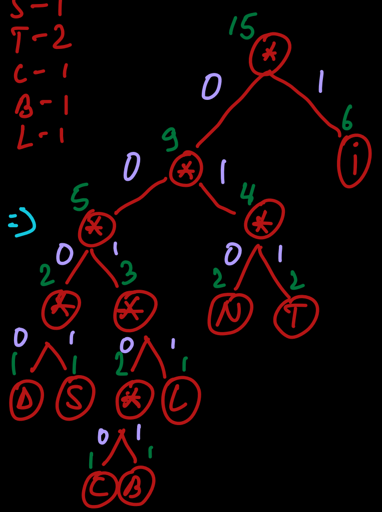
b.

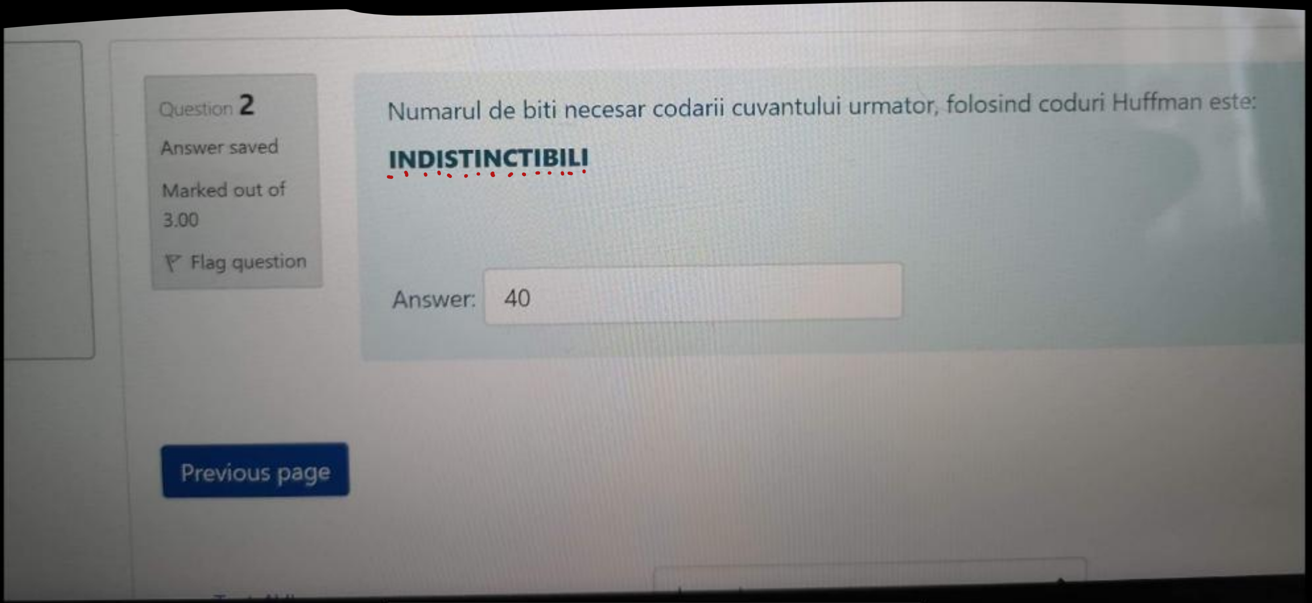


c.

i - 6
N - 2
D - 1
S - 1
T - 2
C - 1
B - 1
L - 1







$\Rightarrow 1*6 + 2*3 + 2*3 + 4 + 4 + 4 + 5 + 5$

$= 6 + 6 + 6 + 4 + 4 + 4 + 5 + 5$

$= 18 + 12 + 10 = 40$

Question 1

Not yet answered

Marked out of 1.00

Flag question

Care din urmatorii arbori Huffman codeaza corect cuvantul de mai jos?

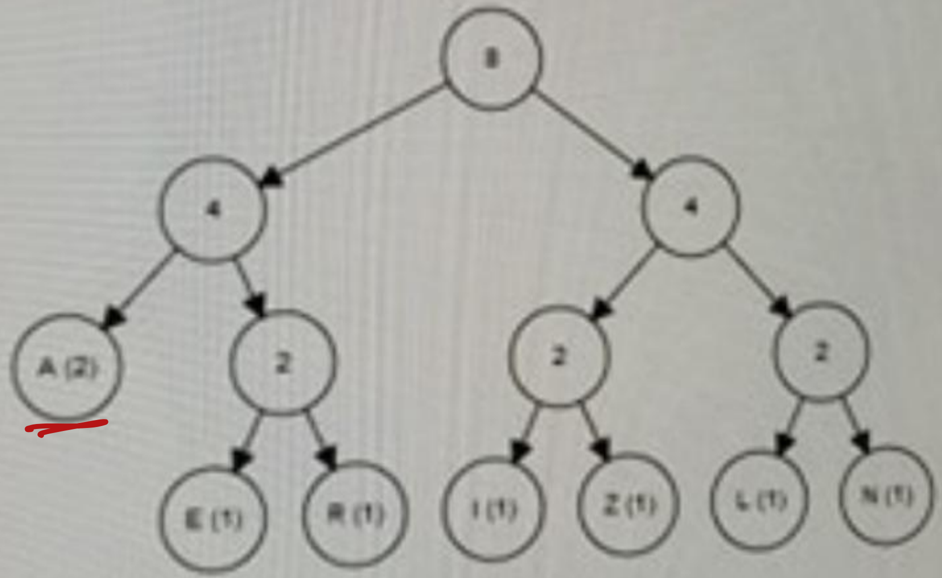
ANALIZARE

ANALIZARE

A - 3
N - 1
L - 1
i - 1
Z - 1
R - 1
E - 1

Select one:

a.



.ro/mod/quiz/attempt.php?attempt=80354&cmid=42278

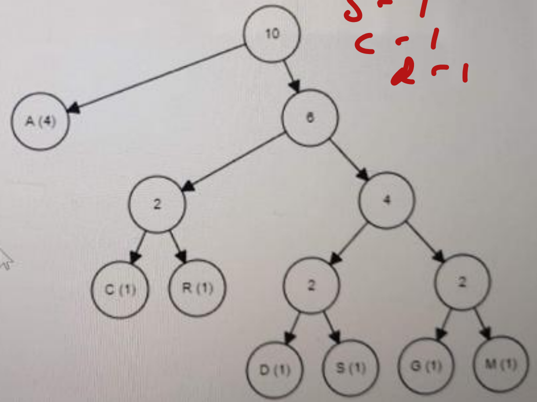
Question 1
Answer saved
Marked out of 3.00
Flag question

Care din urmatorii arbori Huffman codeaza corect cuvantul de mai jos?

MADAGASCAR

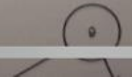
M - 1
A - 5
D - 1
G - 1
S - 1
C - 1
R - 1

Select one:



V

a.



Ancuta

200608431_945625202936431_5581608163752590029_n.jpg

Un cod prefix (care are proprietatea de prefix) poate fi reprezentat printr-un arbore binar, dacă nodurilor terminale ale arborelui li se asociază caracterele originale ale alfabetului, iar ramurilor arborelui ponderile 0 sau 1.

✓

ilor

Care din urmatorii arbori Huffman codeaza corect cuvantul de mai jos?

COTCODACIT

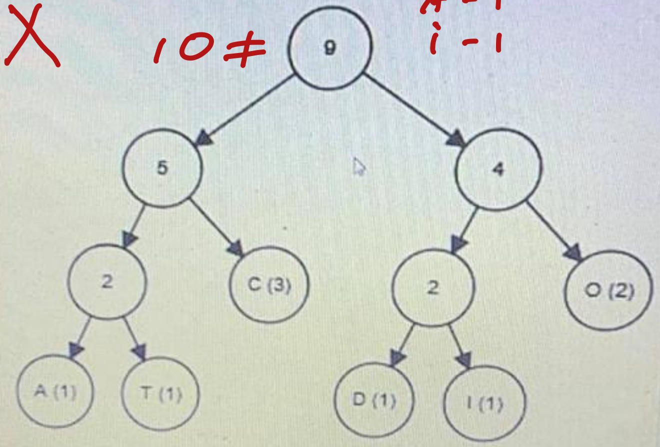
C - 3
O - 2
T - 2
D - 1
A - 1
i - 1

Alegeti o optiune:

X

10 ≠

a.



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A-6
G-1
R-1
M-1
T-3
I-2
C-1
L-1
E-1

GR-2
CM-2
LE-2
GRCM-4
LEI-4
TLEI-7
AGRCM-10
final

A G R M T I C L E

$2\cdot 6+4+4+4+6+6+4+4+4 = \boxed{58}$

Numarul de biti necesar codarii cuvantului urmator, folosind coduri Huffman este:

AGRAMATICALITATEA

Răspuns: 48

dentă

Question **3**

Not yet answered

Marked out of 3.00

Flag question

Numarul de biti necesar codarii cuvantului urmator, folosind coduri Huffman este:

AGRAMATICALITATEA

Answer: 4 8

199252560_124178776456008_5133128136371500654_n.jpg

Question **4**

Not yet answered

Marked out of 2.00

Flag question

Care din urmatoarele afirmatii despre codurile Huffman este adevarata:

Select one or more:

- [ ] a. Algoritmii de tip Huffman static au dezavantajul că necesită cunoaşterea prealabilă a frecventelor de aparitie pentru fiecare simbol din sursa.
- [ ] b. Codarea Huffman este folosita pentru compresie.
- [ ] c. Codarea Huffman poate prezenta pierderi (de informatie), in anumite cazuri.
- [ ] d. In codarea Huffman, niciun cod nu este prefixul altui cod.

Previous page

← Test AVL

Jump to...

Question **3**

Not yet answered

Marked out of 3.00

Flag question

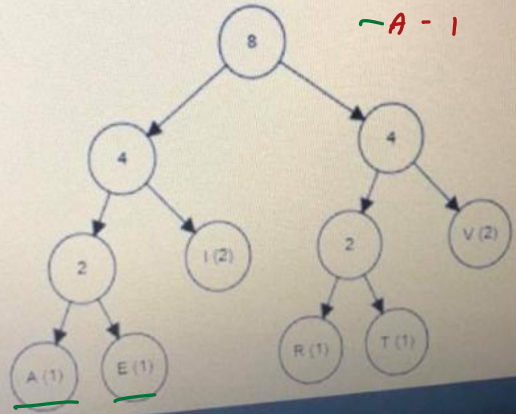
Care din urmatorii arbori Huffman codeaza corect cuvantul de mai jos?

VEVERITA

V - 1
E - 2
R - 1
i - 1
T - 1
A - 1

Select one:

a.



199637502_1205932499906252_3518262501730860748_n.jpg

4 Intrebare

Nu a primit răspuns încă

Marcat din 3,00

Întrebare cu flag

Numarul de biti necesar codarii cuvantului urmator, folosind codul Huffman este:

INDISTINCTIBILI

Răspuns: 40

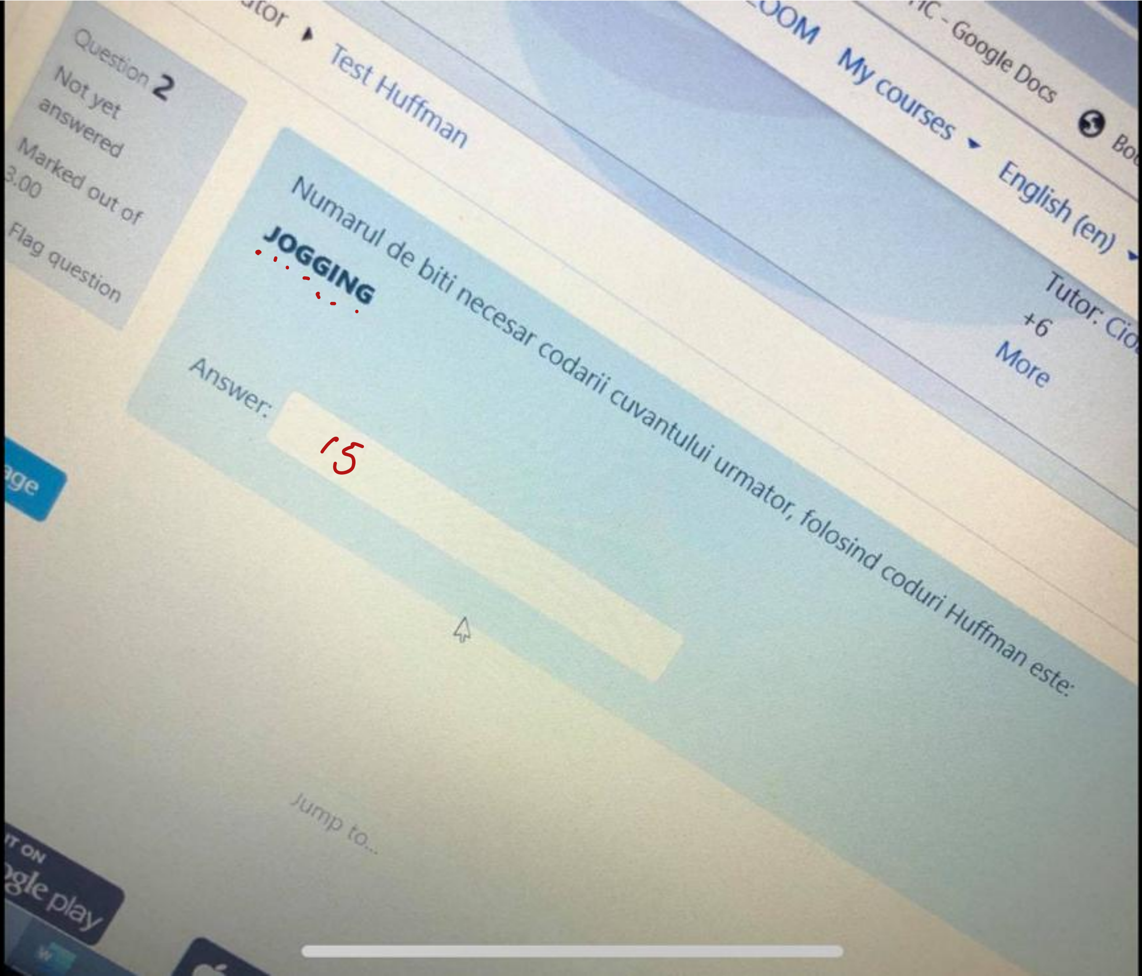
Pagina precedentă

J - 1 }
O - 1
G - 3
i - 1 }
N - 1

=> J,O - 2
G - 3
N,i - 2

=> J,O,N,i - 4
G - 3

=> J,O,N,i,G - 7



J O G i N

J	O	G	i	N
1	1	1	1	1
1	1		1	1
1	1		1	1

=>

3 3 3 3 3 => 15

Test Huffman

Question 1

Not yet answered

Marked out of 2.00

Flag question

Se dorește a se codifica fiecare caracter printr-o secvență de cifre binare "0" și "1", astfel încât codul unui caracter, să nu fie [prefix] pentru codul nici unui alt caracter.

Teachers:

Tutor: CRETU Vladimir - vcard

Tutor: Cioarga Razvan - vcard

+6

More

◄ Test AVL

Jump to...

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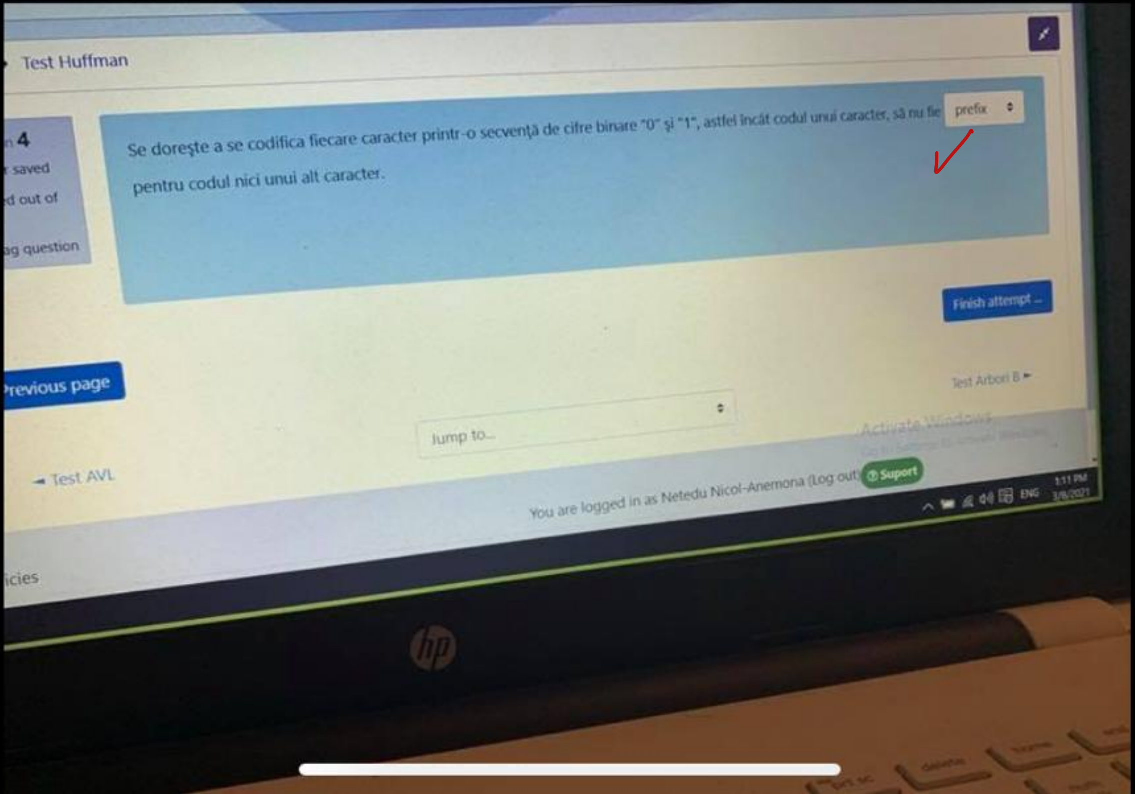
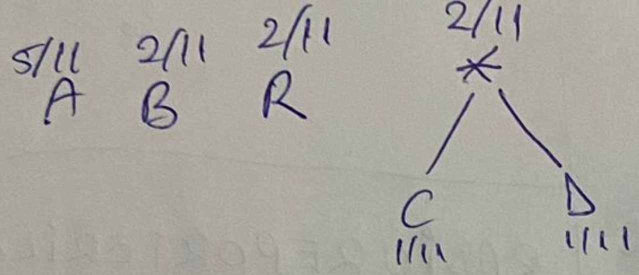
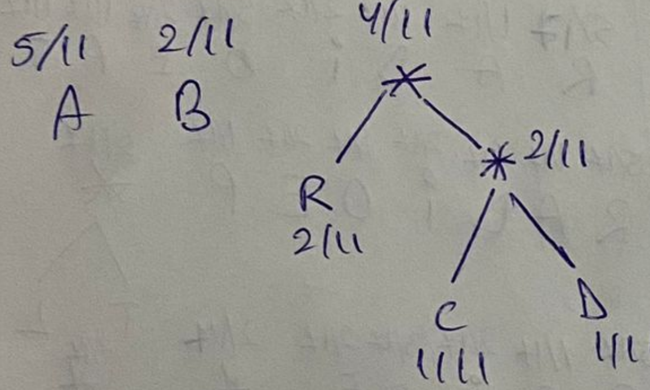
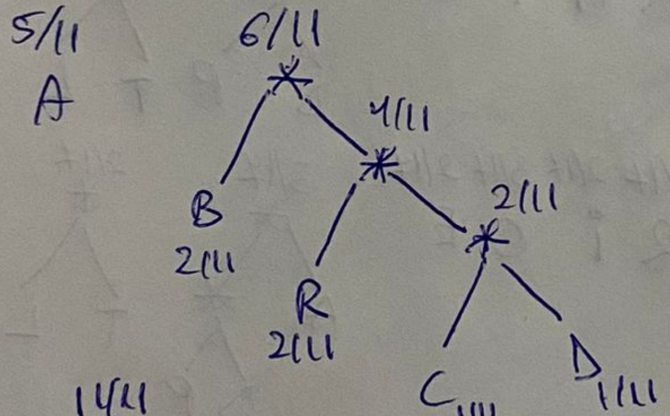


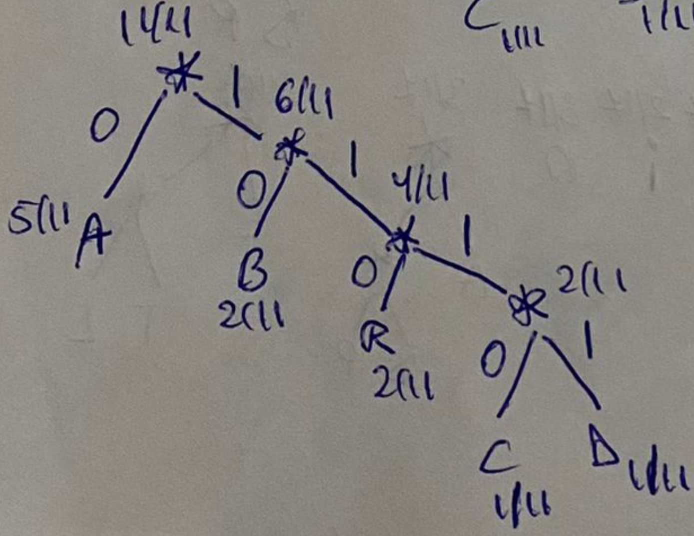
Nr. de biți necesar codării cuv.: ABRACADABRA

5/11	2/11	2/11	1/11	1/11
A	B	R	C	D









A – 0

B – 10

R – 110

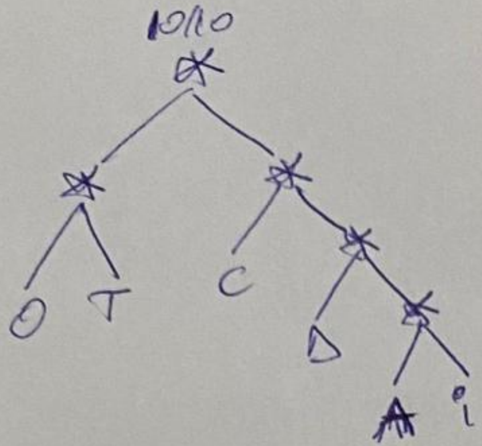
C – 1110

D – 1111

$\underbrace{\text{ABRACADABRA}}_{11}$

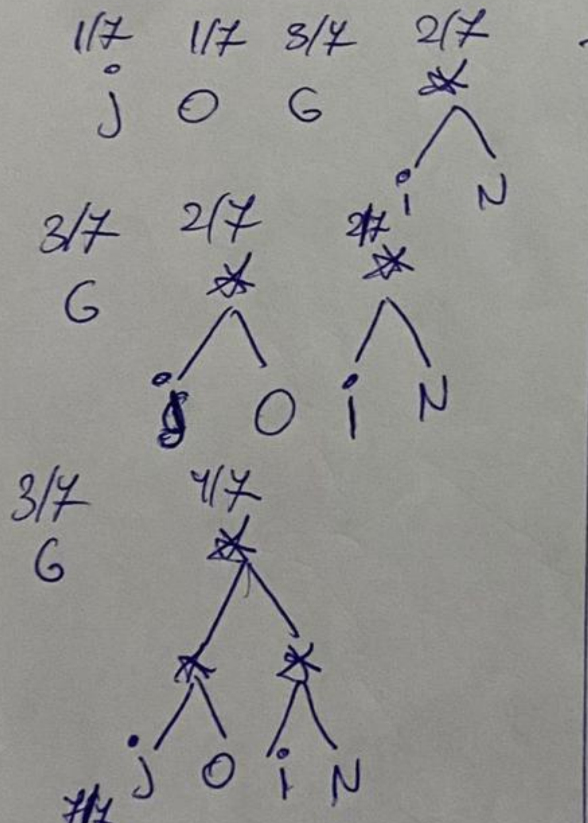
01011001110011110101100

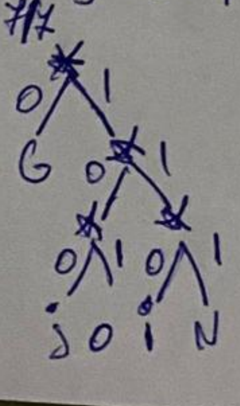
23 de biți



JOGGING

1/7	1/7	3/7	1/7	1/7
J	O	G	I	N



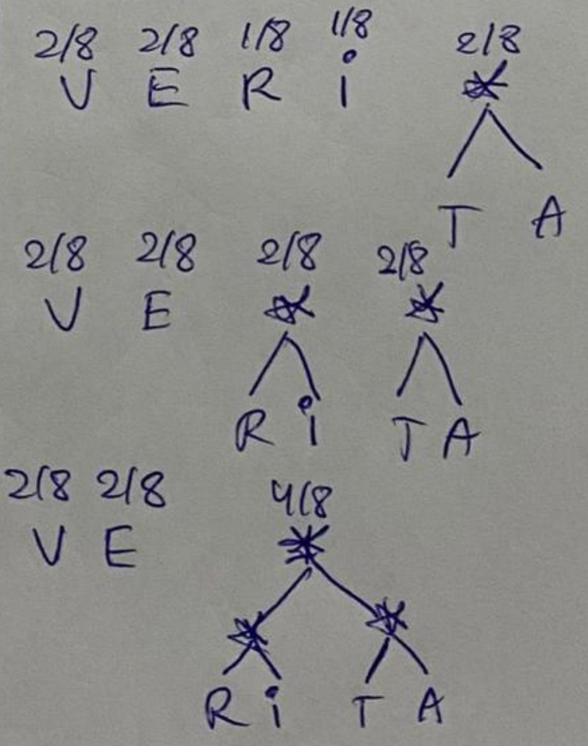


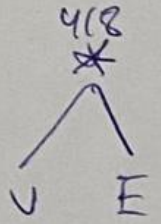
J-100
G-0
O-101
I-110
N-111

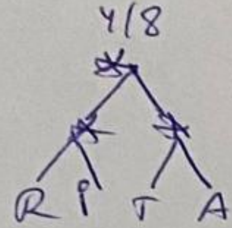
JOGGING-1·3+1·3+3·1+1·3+1·3
+1·3=15

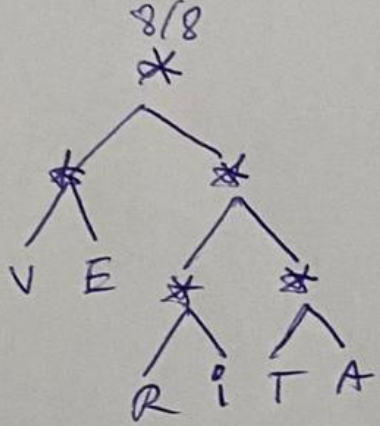
VEVERITA

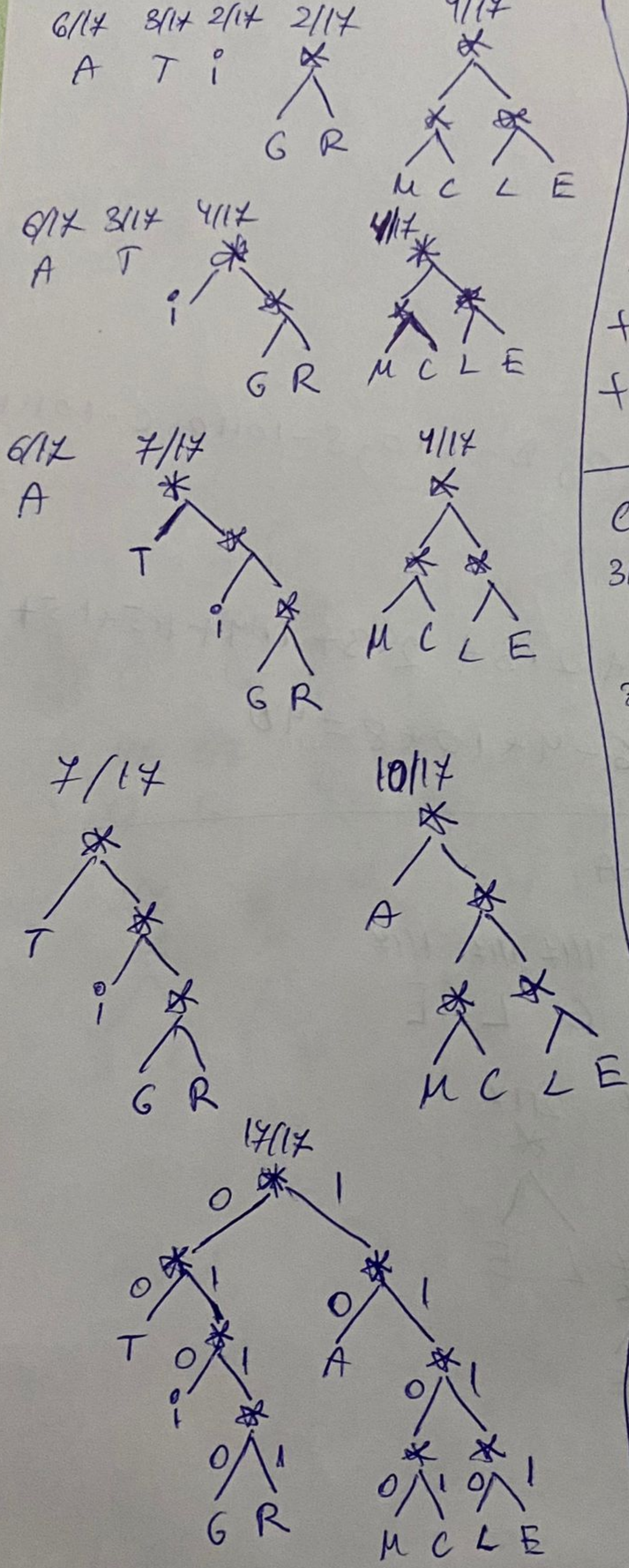
2/8	2/8	1/8	1/8	1/8	1/8
V	E	R	I	T	A




4/8
V
E


4/8
R
i
T
A


8/8
V
E
R
i
T
A



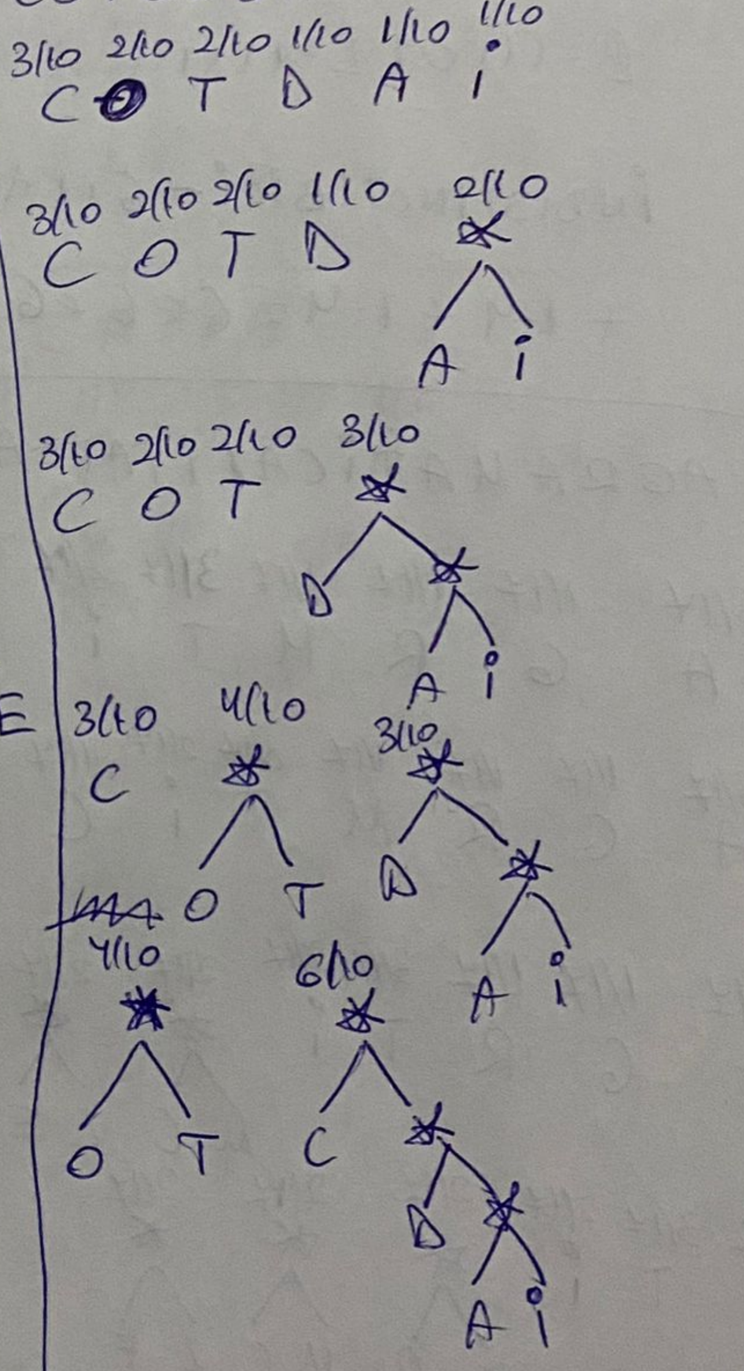
T-00, i-010, G-0110,

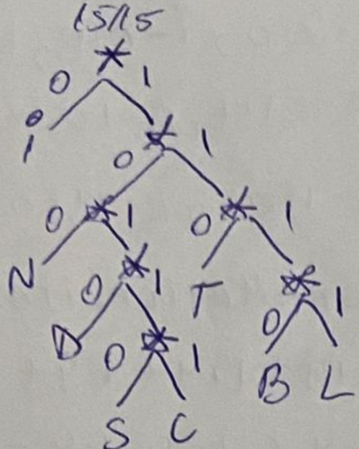
R-0111, A-10, M-1100,

C-1101, L-1110, E-1111

AGRAMATICALITATEA $- 6\cdot2 + 1\cdot4 + 1\cdot4 + 1\cdot4 + 3\cdot2 + 2\cdot3 + 1\cdot4 + 1\cdot4 + 1\cdot4 = 48$

COTCODACIT





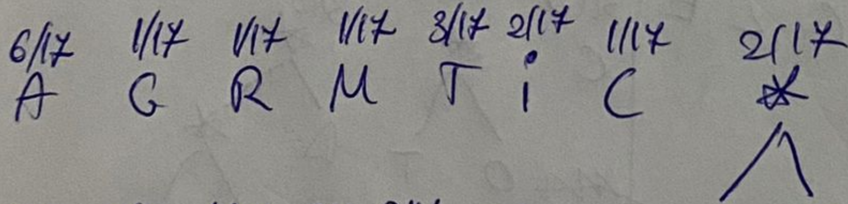
i-0, N-100, T-110, D-1010, S-10110, C-10111

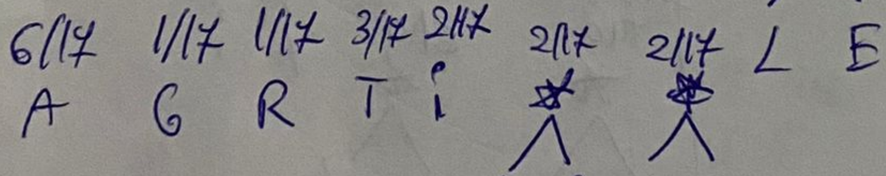
B-1110, L-1111

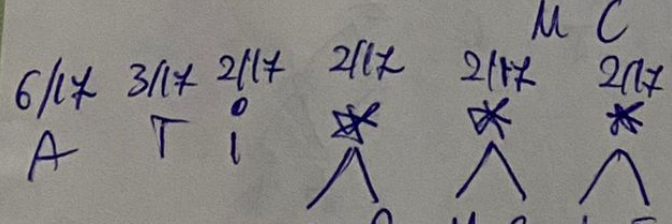
INDISTINCTIBILI - 6·1 + 2·3 + 2·3 + 1·4 + 1·5 + 1·5 + 1·4 + 1·4 = 6 + 6 + 6 + 4 + 10 + 8 = 40

AGRAMATICALITATEA

A	G	R	M	T	i	C	L	E
6/17	1/17	1/17	1/17	3/17	2/17	1/17	1/17	1/17

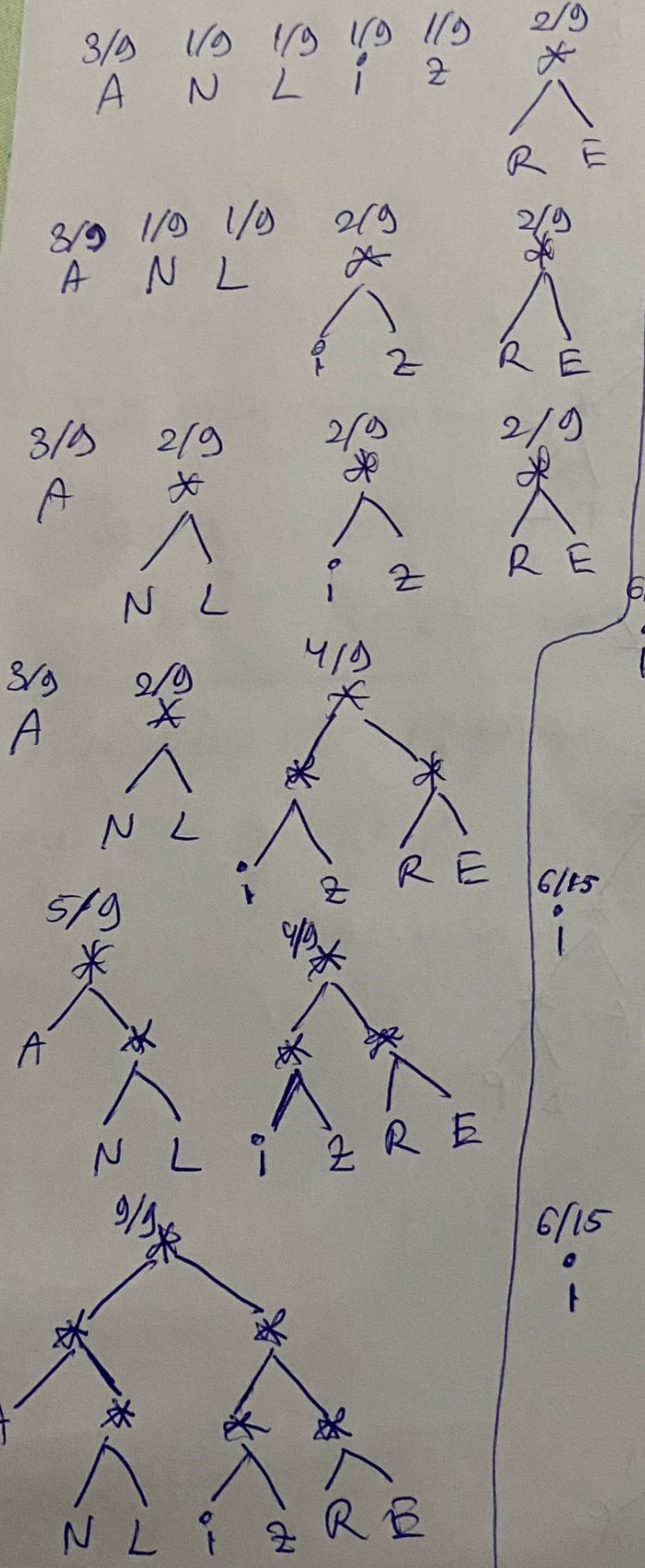






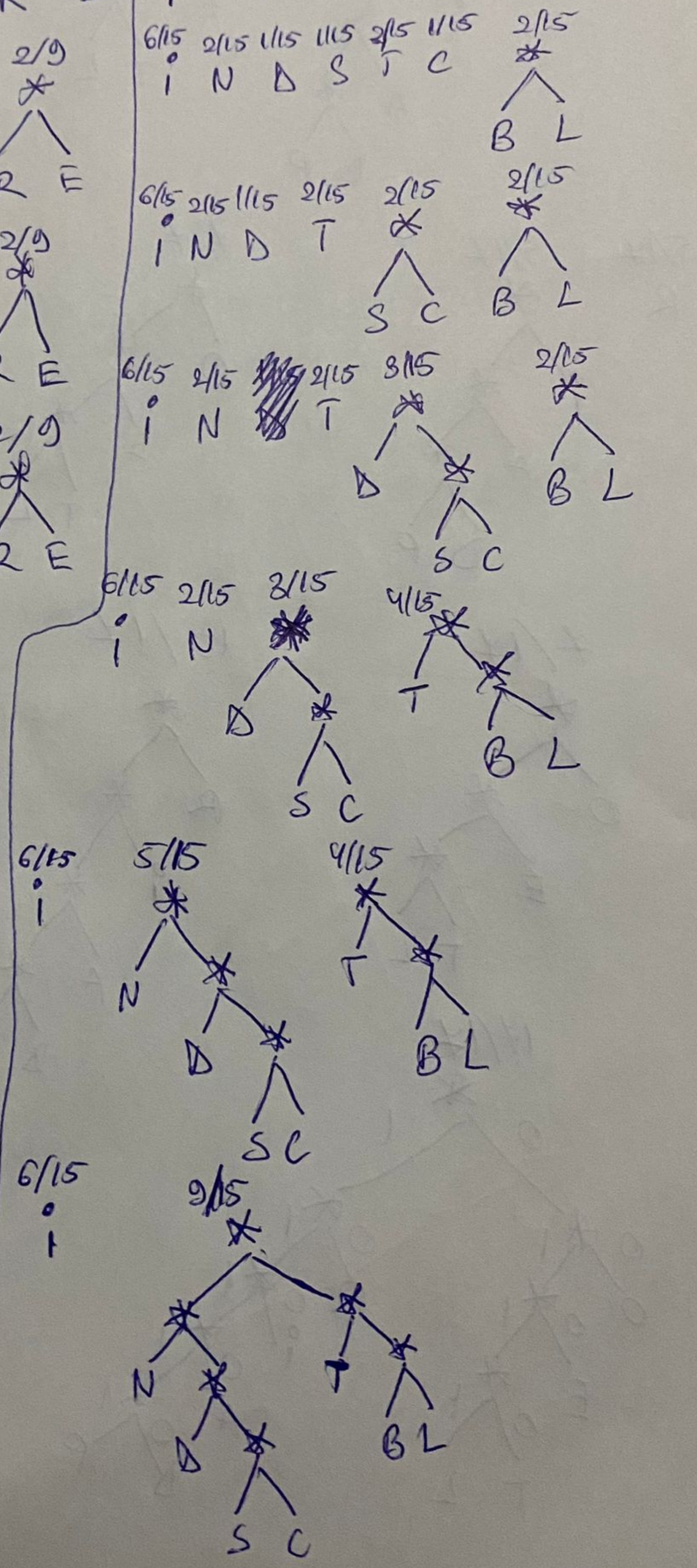
ANALIZARE

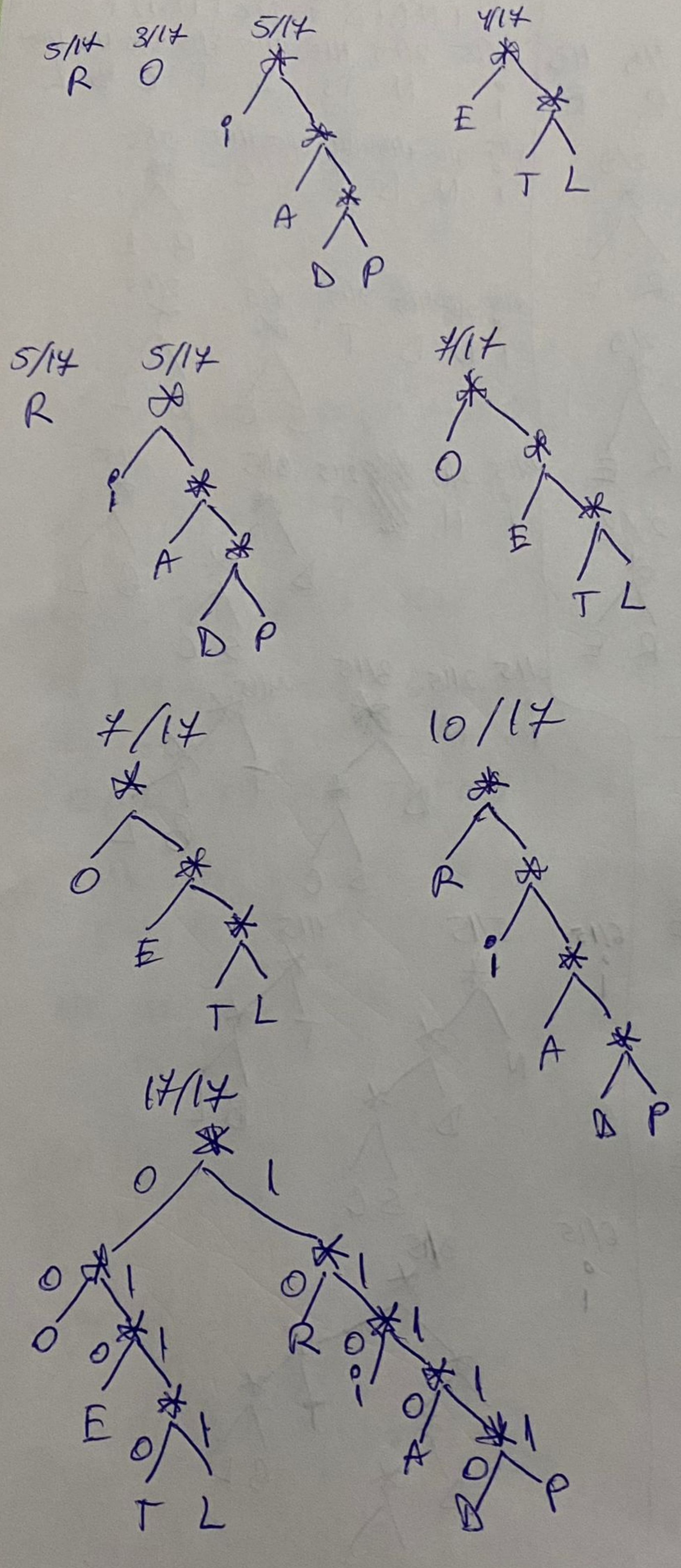
3/9 1/9 1/9 1/9 1/9 1/9 1/9
A N L i Z R E



INDISTINCTIBILI

6/15 2/15 1/15 1/15 2/15 1/15 1/15 1/15
i N D S T C B L



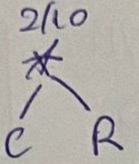


5/17
R
3/17
O
5/17
i
A
D P
4/17
E
T L
5/17
R
5/17
i
A
D P
7/17
O
E
T L
7/17
O
E
T L
10/17
R
i
A
D P
17/17
0 1
O
E
T L
R
i
A
D
P

MADAGASCAR

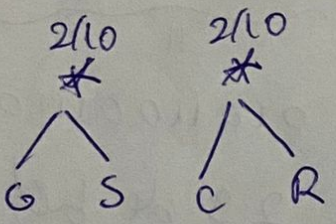
1/10 4/10 1/10 1/10 1/10 1/10 1/10
M A D G S C R

1/10 4/10 1/10 1/10 1/10
M A D G S

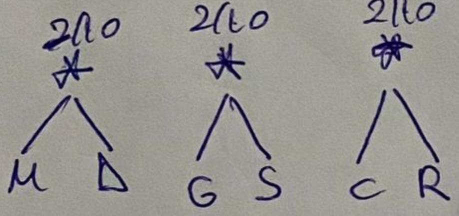


~~1/10 4/10 1/10 1/10~~
~~M A D G~~

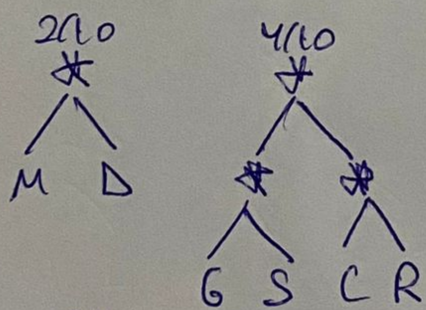
1/10 4/10 1/10
M A D



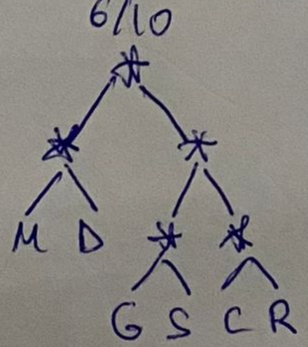
4/10
A

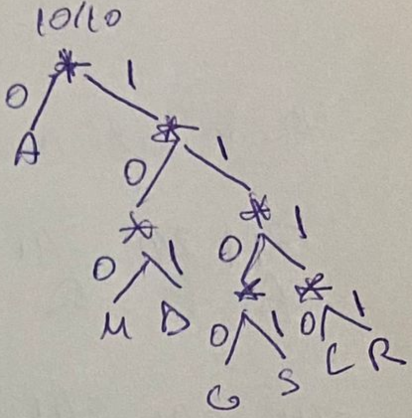


4/10
A



4/10
A

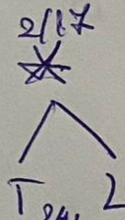




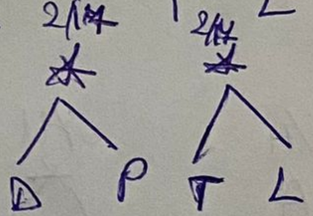
RADIOREPORTERILOR

5/17 1/17 1/17 2/17 3/17 2/17 1/17 1/17 1/17
R A D i O E P T L

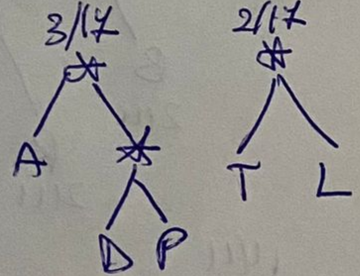
5/17 1/17 1/17 2/17 3/17 2/17 1/17
R A D i O E P



5/17 1/17 2/17 3/17 2/17
R A i O E



5/17 2/17 3/17 2/17
R i O E



5/17 2/17 3/17
R i O